

WINCHESTER, VA, USA

WMO: 724053

Lat: 39.143N

Lon: 78.144W

Elev: 222

StdP: 98.69

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 03717

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-7.9	-19.7	0.7	-7.6	-16.3	0.9	-6.0	12.0	0.9	11.1	0.5	3.1	300	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	33.9	23.4	32.5	23.0	31.2	22.6	25.1	31.5	24.4	30.2	23.7	29.0	3.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.2	27.4	22.6	17.8	26.9	22.3	17.4	26.6	77.5	31.5	74.5	30.0	71.8	28.9	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	8.6	7.5	-14.7	36.6	2.6	3.5	-16.6	39.1	-18.2	41.1	-19.6	43.1	-21.6	45.7		
			-15.4	26.3	2.5	0.9	-17.2	27.0	-18.6	27.5	-20.0	28.0	-21.8	28.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	1.3	2.7	6.8	12.6	17.7	22.2	24.6	23.7	19.9	13.5	7.6	3.1	(6)
(7)		DBStd	9.37	5.74	5.34	5.39	4.86	4.41	3.42	2.94	2.84	3.74	4.57	4.79	4.74	(7)
(8)		HDD10.0	987	278	211	131	26	1	0	0	0	17	102	222		(8)
(9)		HDD18.3	2566	528	438	359	181	68	8	1	1	26	162	322	474	(9)
(10)		CDD10.0	2092	8	7	31	105	239	367	452	425	297	126	30	7	(10)
(11)		CDD18.3	631	0	0	2	10	47	124	194	168	73	13	1	0	(11)
(12)		CDH23.3	5821	0	2	18	126	445	1122	1870	1505	618	110	5	1	(12)
(13)		CDH26.7	2169	0	0	4	29	136	404	793	575	205	22	1	0	(13)
(14)	Wind	WSAvg	2.5	3.1	3.1	3.3	3.1	2.3	2.2	2.0	1.8	1.8	2.3	2.6	2.7	(14)
(15)	Precipitation	PrecAvg	1030	69	57	93	82	95	102	91	84	112	79	78	76	(15)
(16)		PrecMax	1467	180	147	209	159	196	203	169	153	265	194	186	135	(16)
(17)		PrecMin	810	26	13	0	36	29	29	26	34	6	7	11	29	(17)
(18)		PrecStd	177	40	34	44	33	45	47	39	32	74	50	42	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	22.0	26.1	30.1	32.5	34.9	37.2	35.4	33.7	29.6	23.0	20.2	(19)
(20)			MCWB	14.9	14.1	15.9	18.0	22.3	23.7	24.0	24.0	21.8	20.9	15.8	15.0	(20)
(21)		2%	DB	16.3	17.2	22.1	27.1	30.1	32.4	34.6	32.9	31.3	26.5	20.8	16.2	(21)
(22)			MCWB	12.8	11.2	14.0	17.1	21.2	23.3	23.8	23.5	21.7	19.0	14.6	12.2	(22)
(23)		5%	DB	12.9	13.8	18.7	24.0	27.7	31.1	32.5	32.0	28.9	23.6	17.6	12.7	(23)
(24)			MCWB	9.5	8.7	12.5	15.5	19.8	22.8	23.3	23.0	20.8	17.4	13.3	9.3	(24)
(25)		10%	DB	9.8	11.0	16.2	21.6	26.0	29.0	31.1	30.0	27.2	21.3	16.0	10.2	(25)
(26)			MCWB	6.7	7.0	10.6	14.4	18.8	21.8	23.1	22.4	19.9	16.7	11.5	7.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	15.0	17.3	20.5	23.9	25.3	26.1	26.0	24.2	22.2	18.1	16.0	(27)
(28)			MDCB	17.9	19.7	24.0	27.0	30.2	31.6	32.3	32.2	29.3	26.7	21.5	18.6	(28)
(29)		2%	WB	13.5	12.0	15.4	18.4	22.4	24.4	25.2	25.0	23.1	20.6	16.3	12.9	(29)
(30)			MDCB	16.2	16.0	19.7	24.1	28.1	30.4	31.8	31.0	28.1	24.1	18.9	15.5	(30)
(31)		5%	WB	10.1	9.2	13.5	16.9	21.2	23.5	24.5	24.1	22.2	19.0	14.2	10.0	(31)
(32)			MDCB	13.0	13.3	18.0	22.5	26.0	29.1	30.6	29.2	26.8	22.7	17.2	12.4	(32)
(33)		10%	WB	6.8	7.1	11.3	15.4	19.9	22.7	23.7	23.3	21.3	17.3	12.1	7.4	(33)
(34)			MDCB	9.2	10.1	15.7	20.5	24.4	27.6	29.2	28.0	24.9	20.0	14.8	9.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	9.3	10.5	11.9	11.5	11.2	11.3	11.2	11.5	11.2	10.0	8.3	(35)
(36)			MCDBR	12.3	14.3	15.2	15.8	14.2	12.9	13.1	13.1	14.4	14.5	13.4	12.2	(36)
(37)			MCWBR	9.4	9.9	9.2	8.5	7.0	5.6	4.9	5.1	6.0	7.9	9.0	9.3	(37)
(38)		5% WB	MCDBR	11.5	13.3	13.5	13.5	12.6	11.6	11.3	11.0	11.4	12.4	11.6	11.3	(38)
(39)			MCWBR	9.5	9.8	9.1	8.1	6.7	5.5	4.8	4.8	5.6	7.5	8.8	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.317	0.351	0.384	0.421	0.486	0.504	0.501	0.418	0.360	0.318	0.303	(40)	
(41)		taud	2.473	2.406	2.380	2.317	2.236	2.085	2.059	2.046	2.262	2.423	2.504	2.499	(41)	
(42)		Ebn at Noon	876	906	905	892	861	803	784	774	823	844	849	846	(42)	
(43)		Edh at Noon	79	96	110	125	138	161	164	162	123	94	76	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.99	2.83	3.78	4.81	5.35	5.93	5.79	5.19	4.26	3.18	2.28	1.73	(44)	
(45)		RadStd	0.17	0.34	0.22	0.39	0.52	0.44	0.33	0.27	0.54	0.36	0.20	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	-0.74	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+124	+68

Nomenclature: See separate page